

Krutarth Haldankar

Software Engineer

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Summary

Software Engineer focused on building production web applications with React, TypeScript, Angular, and Node.js. Experienced in modernizing legacy systems, translating complex business requirements into maintainable workflows, and working across frontend, APIs, and PostgreSQL. Strong background in state management, performance optimization, authentication, and building reusable full-stack features.

Work Experience

Goa Electronics Limited (GEL)

Feb 2026 – Present

Software Developer

- Developed React and TypeScript features for the modernization of citizen-facing government services, translating legacy application behavior and functional requirements into new frontend workflows.
- Analyzed legacy service implementations, FRS documentation, and UAT behavior to identify discrepancies in validation, field behavior, document requirements, and workflow rules.
- Implemented PostgreSQL stored procedures powering service workflows, including multi-join data retrieval, conditional business logic, dynamic INSERT operations, and data migration between legacy and modernized modules.
- Implemented OAuth-based SSO by handling identity-provider redirects, access-token extraction, and session-token generation with user metadata for authenticated service access.

Ciklum

Jul 2024 – Oct 2025

Software Engineer – Henkel IoT Device Monitoring Dashboard (Angular)

- Built a config-driven multi-step onboarding form reused across three flows – customer registration, sensor addition, and replacement – with dynamic rendering and validation, eliminating duplicate components.
- Implemented global NgRx caching with encrypted localStorage persistence, fetching shared data once and re-fetching only after relevant mutations to avoid redundant API calls across routes.
- Identified the Plotly-based 3D graph as a major bundle contributor; implemented Angular route-level lazy loading to split the analytics module into a separate chunk loaded only when the graph page was accessed.
- Built IoT threshold-breach notifications using NgRx state to track notification visibility and synchronize status with the API only when required.
- Developed plant and sensor listings with search/filtering, optimizing component updates and API calls to respond only to relevant filter-state changes.

Technical Projects

Writing & Diagramming Workspace

React, TypeScript, Node.js, Express, PostgreSQL, Prisma, Tiptap, Excalidraw

- Single-user document workspace with Tiptap rich text editor and embedded Excalidraw canvas — toggle between writing and diagramming mode per document, both states persisted independently in a single structured JSON field.
- Autosave with 2-second debounce and a PATCH delta endpoint — version integer on each document enforces optimistic concurrency control, returning 409 on stale writes to prevent silent data loss across tabs.
- Working draft persisted to localStorage via Redux Persist, synced to backend every 30 seconds and on tab close — users never lose unsaved work between sessions.
- Document version history — stores last 10 snapshots in a DocumentVersion table with automatic pruning, viewable in UI.
- Auth built from scratch: OTP email verification on signup (bcrypt-hashed, 5-minute expiry), JWT access and refresh tokens with silent refresh via Axios interceptor request queue to prevent parallel refresh race conditions.

Skills

- **Languages:** JavaScript, TypeScript
- **Frontend:** React, Next.js, Angular, Tailwind CSS, ShadCN, Material UI, Redux, NgRx, React Query, React Hook Form, Tiptap
- **Backend:** Node.js, Express.js, Prisma ORM, JWT Auth, OAuth, REST APIs
- **Databases:** PostgreSQL, MySQL
- **Tools:** Git, GitHub, Docker (basics), Vercel, Railway, Neon, Cloudinary

Education

Goa College of Engineering

Aug 2020 – Jul 2024

Bachelor of Engineering – Computer Engineering

CGPA: 7.8